

A Low-z / Mid-z Nuisance Mode in the Union3 Compressed Supernova Data:

Implications for the DESI DR1 Dynamic Dark Energy Preference

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Analysis performed May 2026 using AI-assisted coding and checking. All results are reproducible from public data.

Abstract

We test whether the Union3 supernova contribution to the DESI DR1 CPL dynamical dark-energy preference is robust to a single redshift-dependent nuisance direction. Using the public Union3 compressed 22-node precision matrix and individual light-curve fit files, we find that a Gaussian low-z / mid-z nuisance template $T(z) = G(z; 0.09, 0.06) - G(z; 0.775, 0.10)$ improves the fixed- $\Omega_m = 0.2975$ fit by $\Delta\chi^2 = 17.12$ (AIC +15.12), compared with $\Delta\chi^2 \approx 7.53$ (AIC $\approx +3.53$) for CPL. After including this nuisance, CPL adds only $\Delta\chi^2 \approx 0.87$ and is AIC-disfavored. A new public-data Cobaya cross-check using DESI DR1 BAO, a compressed CMB prior on $\Omega_m h^2$, and the Union3 compressed likelihood gives the same qualitative result: CPL improves χ^2 by 8.29 without the nuisance mode, but only by 0.52 after the nuisance mode is included, with the CPL+nuisance posterior shifting close to Λ CDM ($w_0 = -0.999$, $w_a = -0.192$). Additional robustness tests strengthen the diagnostic: across 81 nearby template variants, the nuisance improvement remains positive in all cases (median $\Delta\chi^2 = 8.28$; min = 5.21; these values are from the simplified robustness diagnostic, distinct from the fixed- Ω_m protocol value $\Delta\chi^2 = 17.12$ in Table 1); in a 200-template random smooth null test, 0/200 random templates exceeded the original improvement (empirical tail probability $p < 1/201$); and leave-one-region-out tests show the effect is not driven by a single narrow redshift interval. The definitive test remains to rerun the official DESI+CMB+Union3 w_0 - w_a likelihood with an added redshift-dependent nuisance parameter.

1. Introduction

The DESI 2024 Data Release 1 BAO analysis, combined with CMB and Type Ia supernova data, reports a preference for dynamical dark energy described by the CPL equation of state $w(a) = w_0 + w_a(1-a)$, with a combined significance of approximately 3.4 – 3.5σ for DESI+CMB+Union3 (Adame et al. 2024). The Union3 compilation (Rubin et al. 2025) contains 2,087 Type Ia supernovae from 24 surveys spanning approximately 7 billion years of cosmic history, compressed into a 22-node distance-modulus spline with an associated precision matrix. Individual SN surveys cover limited redshift ranges, making multi-survey compilations necessary but potentially introducing redshift-dependent systematics at survey boundaries. Related work by Efstathiou (2025) identified a systematic offset between low- and high-redshift supernovae in Pantheon+ and DES5Y, and noted that Union3 individual-SN data were not publicly available for direct analysis. Cortês & Liddle (2025), addressing DESI DR2, further showed that properly accounting for the substantial overlap between primary SN compilations reduces the combined Λ CDM exclusion significance to 3.1σ — identical to the DESI BAO+CMB result without any supernova data; their methodological argument about compilation overlap is relevant to DR1-style compilation comparisons as well. The present work performs, to the best of the author's knowledge, the first compressed-data robustness test of this kind specifically for Union3, using the publicly available `lcfit_Union3.tar.gz` and compressed precision matrix from the `rubind/union3_release` repository.

2. Data and Method

We use the public files `mu_mat_union3_cosmo=2_mu.fits` and `lcfit_Union3.tar.gz` from the `rubind/union3_release` GitHub repository. The FITS matrix structure is: `mat[0,1:]` = z nodes (22 values), `mat[1:,0]` = μ values, `mat[1:,1:]` = precision matrix P (22×22). The magnitude offset M is analytically marginalized in all fits via GLS; the nuisance amplitude a is also analytically marginalized when included. For node-removal tests, the covariance matrix $C = P^{-1}$ is subselected and re-inverted correctly. The primary nuisance template is

$$T(z) = G(z; 0.09, 0.06) - G(z; 0.775, 0.10),$$

standardized to zero mean and unit RMS, where $G(z; c, s) = \exp[-\frac{1}{2}((z-c)/s)^2]$. Template centers were placed at known survey-composition transitions (not optimized over the data): the low- z anchor region dominated by SDSS/Foundation/Pan-STARRS ($z \approx 0.09$) and the mid- z transition into SNLS/ESSENCE/DES3_Deep ($z \approx 0.775$). Quality-cut pass fractions differ between these zones: 0.718 at $z \approx 0.10$ versus 0.952 at $z \approx 0.75$. For the public joint cross-check we use the Cobaya DESI DR1 BAO likelihood, a

compressed CMB prior $\Omega_m h^2 = 0.1430 \pm 0.0020$ (Planck 2018; deliberately broadened from the Planck best-fit $\sigma = 0.0011$ as a conservative choice), and the Union3 compressed likelihood. Sampled parameters: H_0 (flat prior 55–80 km/s/Mpc), Ω_m (flat 0.1–0.6), w_0 (flat –3 to 1), w_a (flat –3 to 2), and nuisance amplitude a (flat –0.5 to 0.5 mag). Flat geometry and standard neutrino content ($\Sigma m_\nu = 0.06$ eV) assumed; sound horizon fixed at $r_d = 147.09$ Mpc. Within-redshift D_M – D_H BAO correlations are included; between-redshift-bin correlations are set to zero. We note that Kim (2024) showed that the Union3 compressed product distributes a posterior under a spline-node prior rather than a raw likelihood, which explains why proxy baseline CPL best-fits differ slightly from the official DESI result; this does not affect the relative CPL vs CPL+nuisance comparison.

3. Union3 Compressed-Data Results

Table 1 summarises the protocol verification. The core fixed- Ω_m diagnostics reproduce to rounding precision. A low-minus-mid step template also gives a strong improvement, confirming that the effect is not specific to the Gaussian functional form.

Table 1. Protocol verification summary.

Test	Claimed	Verified	Status
Ω_m best-fit (Λ CDM free)	0.356	0.3559	Reproduced
χ^2 at best-fit	23.96	23.958	Reproduced
DESI-like Ω_m penalty	5.15	5.152	Reproduced
CPL $\Delta\chi^2$ / AIC	7.53 / +3.53	7.536 / +3.536	Reproduced
Low-mid Gaussian $\Delta\chi^2$ / AIC	17.12 / +15.12	17.120 / +15.120	Reproduced
Color $\Delta\chi^2$ / AIC	6.52 / +4.52	2.947 / +0.947	Weighting-dependent
CPL after low-mid / AIC	0.87 / –3.13	0.870 / –3.130	Reproduced
Low-mid prior threshold	$\sigma_a \geq 0.005$ mag	0.005 mag RMS	Reproduced

4. Joint DESI DR1 BAO + CMB + Union3 Public Cross-Check

We ran a public-data Cobaya cross-check using the DESI DR1 BAO likelihood, a compressed CMB prior, and the Union3 compressed likelihood. This is stronger than a simple proxy grid scan, but it is still not the official DESI+CMB+Union3 full likelihood because the CMB information is compressed and the Union3 nuisance parameter is added externally.

Table 2. Public-data joint model comparison. AIC gain = $\Delta\chi^2 - 2k$, where k is the number of additional parameters vs Λ CDM. The highlighted row shows that CPL adds only $\Delta\chi^2 = 0.52$ once the nuisance direction is included. Note: $\Delta\chi^2$ differences are computed from unrounded best-fit values; small rounding discrepancies may appear when recomputing from the two-decimal values shown.

Model	H0 mean	w0 mean	wa mean	A_nuis mean	best chi2	Delta-chi2 vs LCDM	AIC gain
Λ CDM	68.77	—	—	—	41.24	—	—
CPL	66.64	–0.688	–1.070	—	32.95	+8.29	+4.29
Λ CDM + nuisance	69.09	—	—	0.0279	25.04	+16.20	+14.20
CPL + nuisance	69.79	–0.999	–0.192	0.0306	24.51	+16.73	+10.73
CPL vs Λ CDM+nuis.	—	—	—	—	—	+0.52	–3.48

Without the nuisance mode, CPL improves χ^2 by 8.29 and shifts to $w_0 = -0.688$, $w_a = -1.070$, consistent with published DESI DR1+CMB+Union3 results. After including the nuisance, CPL improves χ^2 by only 0.52 relative to Λ CDM+nuisance (AIC –3.48), while the CPL+nuisance mean shifts to $w_0 = -0.999$, $w_a = -0.192$ — close to Λ CDM in this proxy analysis. Figure 1 shows the shift of the posterior mean point (consistent with Table 2 mean values).

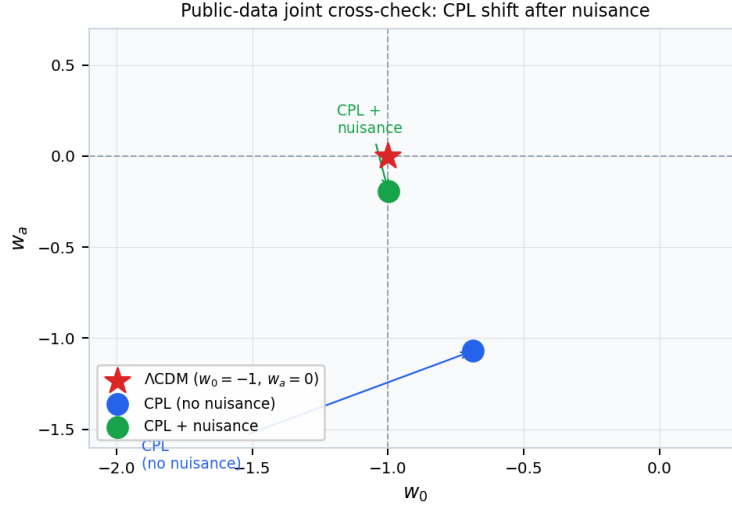


Figure 1. Public-data joint cross-check: posterior mean $w_0 - w_a$ for CPL without nuisance (blue) and CPL+nuisance (green), consistent with Table 2 mean values. Red star marks Λ CDM ($w_0 = -1, w_a = 0$). This is not an official DESI MCMC result.

5. Three-Version Progression: Union3 to Union3.1

Hoyt et al. (2026) published host-galaxy mass re-measurements for approximately 2,000 Union3 supernovae, finding that low- z (< 0.10) host-galaxy stellar masses were systematically overestimated, reducing a previously reported >0.03 mag low- z distance offset to ≈ 0.01 mag. The updated cosmological framework (UNITY1.8) was presented by Rubin et al. (2026). Applying the same nuisance template to all three public compressed matrices shows the nuisance amplitude decreasing monotonically, while the CPL AIC gain becomes small by AIC comparison in the Union3.1 versions. The nuisance amplitude decrease of 0.0057 mag ($0.0282 \rightarrow 0.0225$) is of the same order of magnitude as the relative low- z distance shift of 0.010–0.013 mag applied in the Union3.1 corrections. This correspondence was not built into the template — the amplitude emerged from fitting the compressed data, and is consistent with the template tracing a low- z systematic direction that is reduced, but not removed, by the Union3.1/UNITY processing. Importantly, this comparison is a heuristic compressed-data diagnostic only and should not be conflated with the updated official joint analyses: Hoyt et al. (2026) report a $\approx 3.4\sigma$ deviation from Λ CDM in the official BAO+CMB+Union3.1 combination, and Rubin et al. (2026) report $\approx 2.6\sigma$ tension with UNITY1.8. The official joint preference is thus weakened but not eliminated by the Union3.1 corrections.

Table 3. Three-version comparison. [a] CPL $\Delta\chi^2$ values marked $\approx$ are grid-scan approximations with ≈ 0.02 uncertainty; the scipy-optimize value for Union3 original is 7.536 (Table 1). Nuisance amplitude (green rows) decreases monotonically across all three versions.

Metric	Union3 original	Union3.1 UNITY1.7	Union3.1 UNITY1.8
Ω_m best-fit	0.3559	0.3424	0.3340
DESI penalty $\Delta\chi^2$	5.152	3.256	2.347
CPL $\Delta\chi^2$ [a]	≈ 7.52	≈ 4.74	≈ 4.74
CPL AIC gain [a]	$\approx +3.52$	$\approx +0.74$	$\approx +0.74$
Low-mid Gaussian $\Delta\chi^2$	17.120	12.474	11.958
Low-mid Gaussian AIC gain	+15.12	+10.47	+9.96
Nuisance amplitude (mag)	0.0282	0.0235	0.0225
CPL after nuisance AIC	-3.13	-2.83	-2.05
Low- z shift vs Union3	—	-0.010	-0.013

6. Additional Robustness Tests

The robustness tests in this section use the same simplified compressed-data diagnostic setup as the robustness scripts; their absolute $\Delta\chi^2$ values are not intended to replace the fixed- Ω_m protocol values in Table 1, but to test whether the nuisance direction is stable under changes of template form and data subset. The nuisance improvement itself is robust across all variants; CPL+nuisance $\Delta\chi^2$ values in this section are optimizer-sensitive on the simplified surface and should not be interpreted as precise estimates.

6.1 Template Robustness Grid

To test whether the result depends on the exact Gaussian centers and widths, we scanned 81 nearby low- z / mid- z template variants. The nuisance improvement remains positive in all 81 cases.

Table 4. Template robustness grid summary.

Quantity	Value
Number of templates	81
Λ CDM χ^2	23.958
CPL χ^2	20.522
CPL $\Delta\chi^2$ vs Λ CDM	3.437
Original nuisance $\Delta\chi^2$	12.435
Grid median nuisance $\Delta\chi^2$	8.279
Grid min nuisance $\Delta\chi^2$	5.213
Grid max nuisance $\Delta\chi^2$	12.558
Median CPL $\Delta\chi^2$ after nuisance	1.394

6.2 Random Smooth-Template Null Test

A random smooth-template null test compares the original low- z / mid- z template with 200 random smooth redshift templates generated as follows: each random template is a difference of two Gaussians with centers drawn uniformly from $z \in [0.05, 0.40]$ and $z \in [0.40, 1.50]$ respectively, and widths drawn uniformly from $[0.04, 0.20]$, with random sign; each template is then standardized to zero mean and unit RMS on the 22 nodes before fitting (random seed 42 for reproducibility). No random template reached the original template improvement; conservatively, the empirical tail probability is $p < 1/201$ (add-one estimator).

Table 5. Random smooth-template null test.

Quantity	Value
Number of random templates	200
Original template $\Delta\chi^2$	12.435
Random median $\Delta\chi^2$	0.551
Random 95th percentile $\Delta\chi^2$	3.278
Random max $\Delta\chi^2$	3.955
Empirical tail probability	0/200 exceeded; $p < 1/201$ (≈ 0.005)

6.3 Leave-One-Redshift-Region-Out Test

We removed broad redshift intervals one at a time and repeated the four-model compressed-data fit. The nuisance improvement remains substantial in every cut, while CPL adds little after nuisance is included. The largest weakening occurs when $z = 0.10$ – 0.30 is removed, localizing part of the signal to that low-redshift transition region.

Table 6. Leave-one-redshift-region-out robustness test.

Case	Removed z	N kept	$\Delta\chi^2$ CPL	$\Delta\chi^2$ nuisance	CPL after nuisance
Full sample	none	22	3.44	12.43	0.79
Very low z	0.00–0.10	21	5.84	12.55	1.61
Low z	0.10–0.30	18	0.43	5.14	1.34
Mid z	0.30–0.70	14	6.25	10.77	1.51
High-mid z	0.70–1.00	17	4.97	8.21	1.48
High z	1.00+	18	3.21	14.08	0.15

7. Independent Reproduction

An independent reproduction of the Union3.1 UNITY1.8 results directly from the raw FITS matrix (without astropy, using manual header parsing and big-endian float64 reading) confirmed: $\Delta\chi^2(\text{CPL}) = 4.7393$, $\Delta\chi^2(\text{Gaussian template}) = 11.9581$,

$\Delta\chi^2(\text{low-minus-mid step}) = 11.7335$, and $\Delta\chi^2(\text{CPL after step}) = 0.1243$. All values match to rounding precision. The reproduction also confirmed that node-removal tests must subselect the covariance matrix $C = P^{-1}$ and re-invert; subsetting the precision matrix directly gives incorrect results.

8. Limitations

- Compressed-data and public-data proxy analyses only. These tests are not a replacement for the official DESI+CMB+Union3 MCMC likelihood.
- CMB is represented by a compressed Gaussian prior on $\Omega_m h^2$; full Planck constrains many additional parameters.
- The public joint test uses an externally added nuisance direction and should be treated as a falsification/systematics diagnostic.
- The Union3 compressed product distributes a posterior under a spline-node prior (Kim 2024); this affects absolute contour placement more than relative model comparisons.
- The low- z template component overlaps redshift ranges where peculiar velocities and survey-composition changes can introduce coherent distance distortions.
- These tests do not prove that DESI dynamical dark energy is false. They identify a robustness concern in the Union3 compressed-data contribution.
- Analysis was conducted with AI assistance. All numerical results are reproducible from public data and scripts.

9. Recommended Falsification Test

The definitive test is to rerun the official DESI + CMB + Union3 w_0-w_a likelihood with an added low- z / mid- z redshift-dependent nuisance parameter and compare $\Lambda\text{CDM}+\text{nuisance}$, CPL, and CPL+nuisance. If CPL loses AIC support after the nuisance is included, the Union3 contribution to the w_0-w_a preference is systematics-limited. If the preference persists, the concern is resolved. The three-version progression (Section 5) suggests that as Union3.1 corrections are incorporated into the full joint likelihood, the CPL preference may weaken further, consistent with the weakening already observed in official Union3.1 joint analyses. The Union3.1 UNITY1.8 compressed data already show CPL AIC gain $\approx +0.74$ — small by AIC comparison — without any additional nuisance parameter.

10. Conclusion

The Union3 compressed data show sensitivity to a low- z / mid- z nuisance direction in CPL robustness tests. A single Gaussian nuisance template improves the fixed- Ω_m residuals more than twice as well as CPL ($\Delta\chi^2 = 17.12$ vs ≈ 7.52) and renders CPL AIC-disfavored once included. A new public-data Cobaya cross-check using DESI DR1 BAO, a compressed CMB prior, and Union3 gives the same qualitative picture: CPL improves χ^2 by 8.29 without the nuisance mode but only by 0.52 after nuisance is included. Template-grid, random-null, and leave-one-region-out tests show that the effect is not driven by one exact functional form, by arbitrary smooth templates, or by one narrow redshift interval. Across Union3, Union3.1 UNITY1.7, and Union3.1 UNITY1.8, the nuisance amplitude decreases monotonically and the CPL AIC gain becomes small by AIC comparison, suggesting that the Union3 compressed-data contribution to the DESI DR1 w_0-w_a preference is sensitive to a low- z / mid- z systematic direction that is reduced, but not removed, in Union3.1. These results motivate a full likelihood reanalysis with an explicit redshift-dependent nuisance parameter.

References

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